

# Bursting (beta) rhythms

Instructor: Mark Kramer

# Today

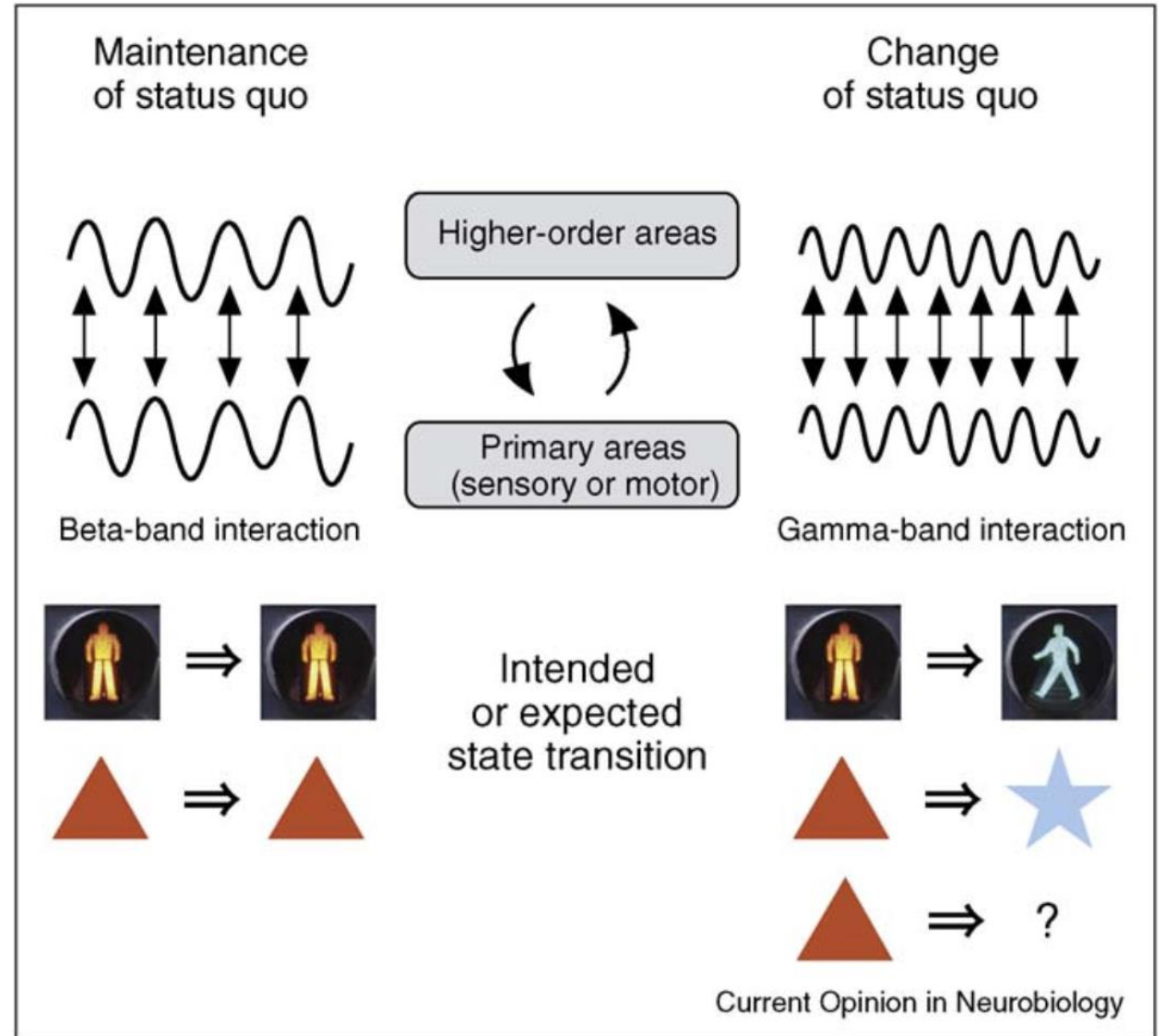
Model of the bursting (beta) rhythm

# Beta rhythms

12-30 Hz

top-down executive control

goal-directed behavior

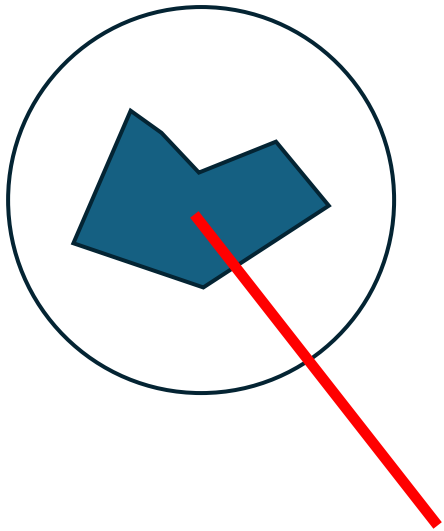


[Engel & Fries, Current Opinion in Neurobio, 2010]

# Beta rhythms

20-30 Hz

experiment

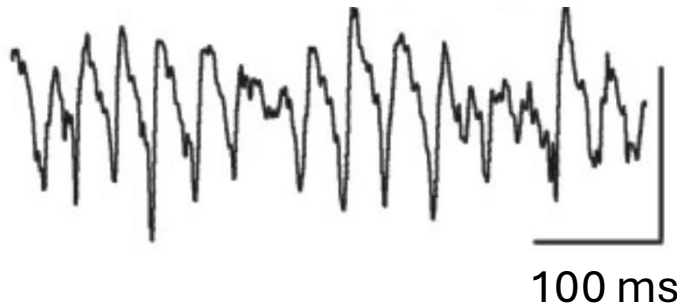


*in vitro* preparation

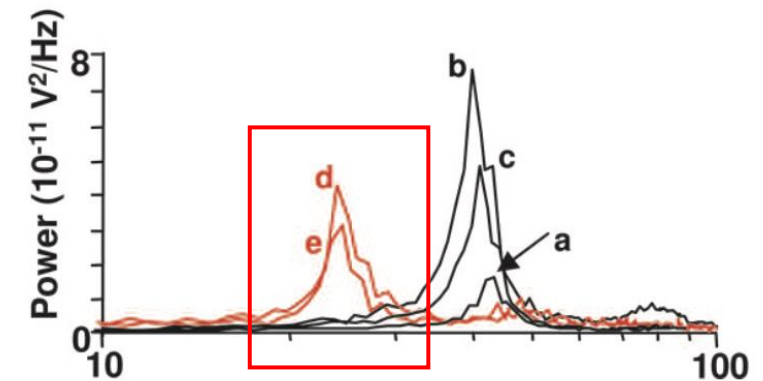
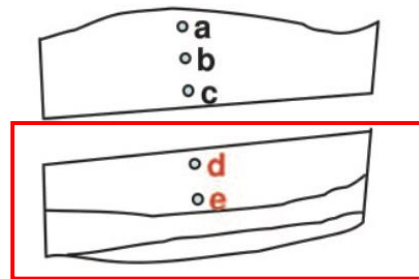
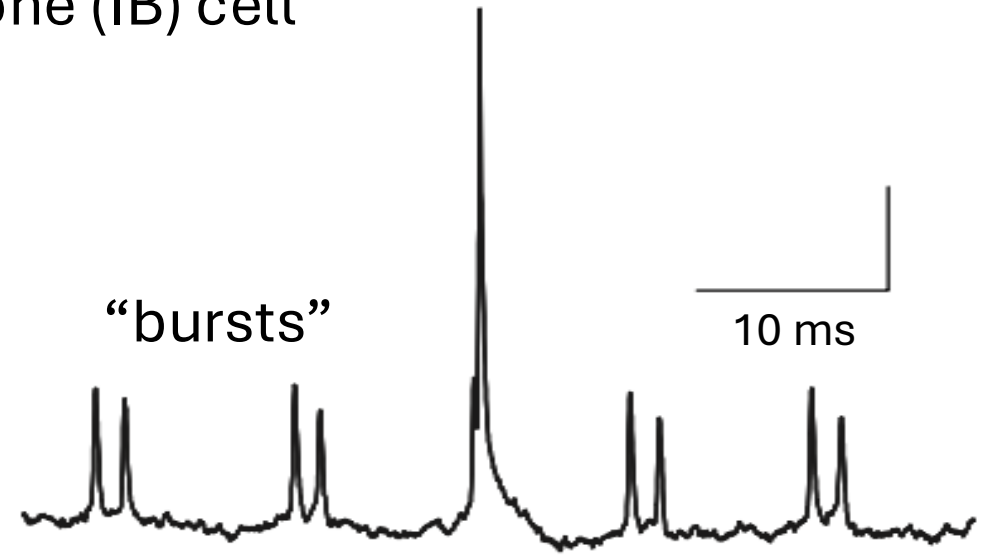
increase excitation (kainate)

block synaptic transmission

record LFP



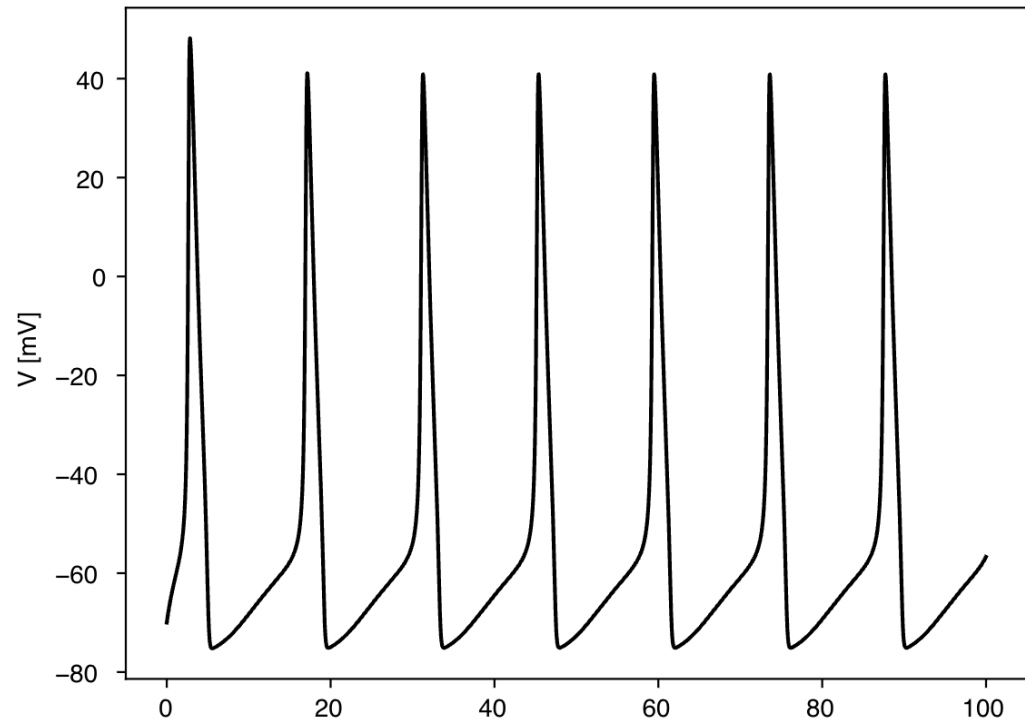
Record from one (IB) cell



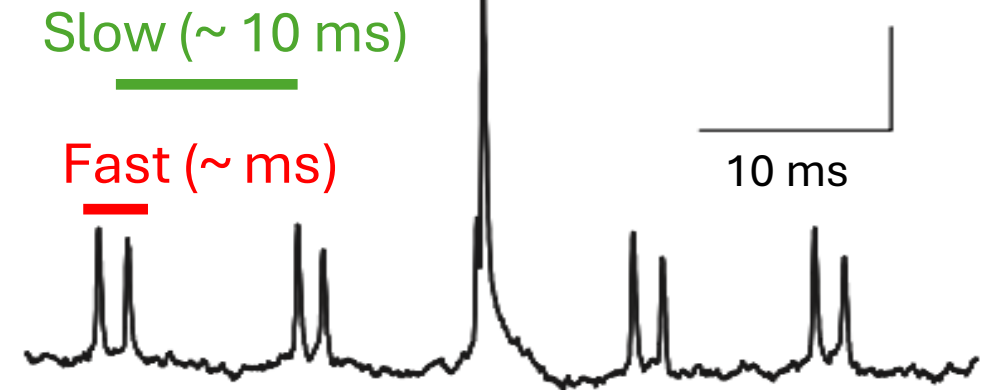
[Roopun et al., PNAS, 2006]

# Beta rhythm model

Our existing models (LIF, HH, ING, PING) do not burst



one timescale



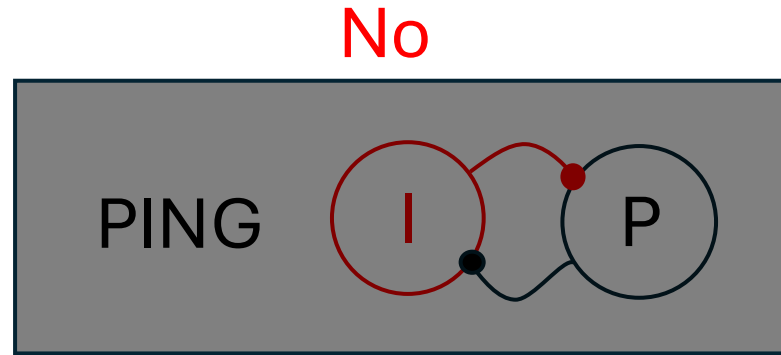
two timescales

**Q.** Model this?

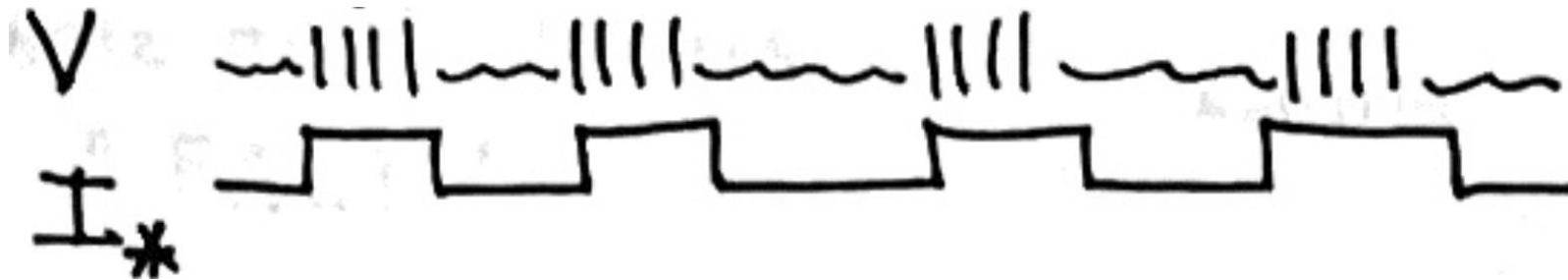
# Beta rhythm model

## Mechanisms

- not synaptic
- another ion current?



**Intuition:** imagine injecting a modulated current  $I_*$  to I&F model



bursting ...  $I_*$  sets the slow timescale

**Idea:** update HH model with a new current that acts like  $I_*$

# Beta rhythm model

Remember HH

$$C \frac{dV}{dt} = I_{\text{input}}(t) - \bar{g}_K n^4 (V - V_K) - \bar{g}_{\text{Na}} m^3 h (V - V_{\text{Na}}) - \bar{g}_L (V - V_L)$$

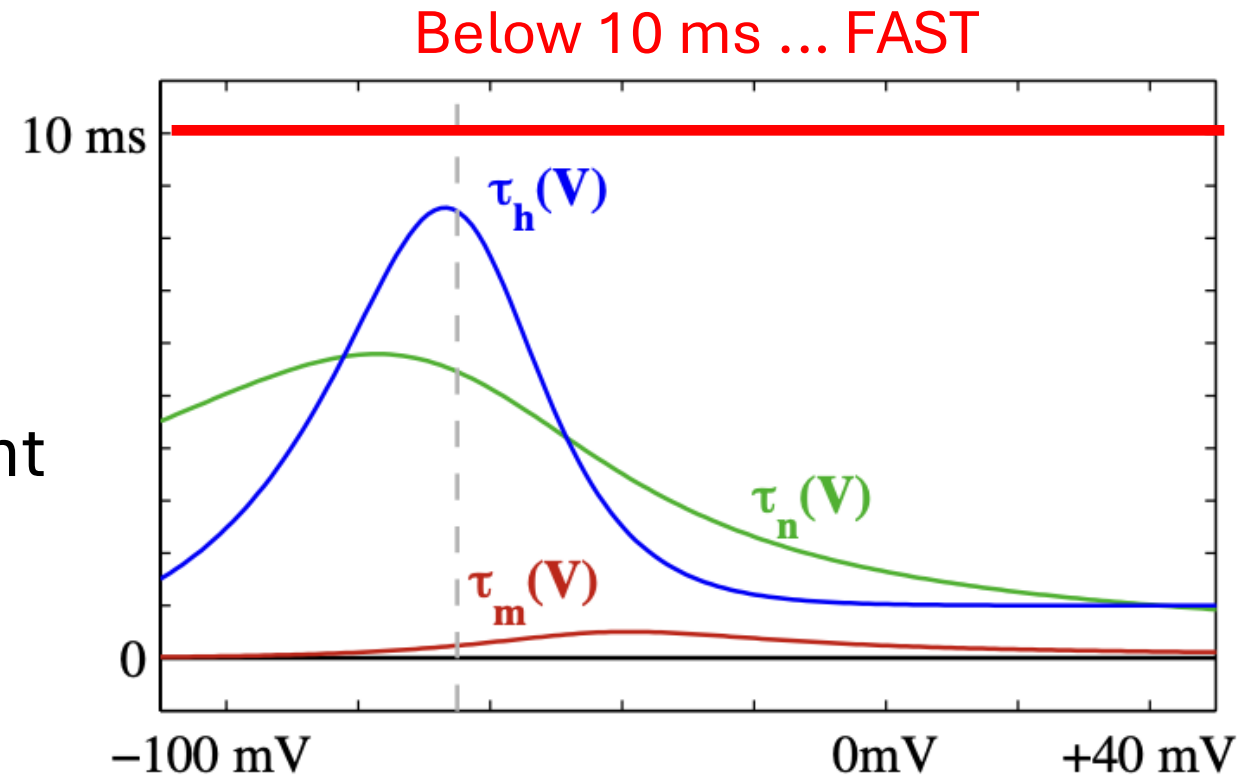
$$\frac{dn}{dt} = - \frac{n - n_{\infty}(V)}{\tau_n(V)}$$
$$\frac{dm}{dt} = - \frac{m - m_{\infty}(V)}{\tau_m(V)}$$
$$\frac{dh}{dt} = - \frac{h - h_{\infty}(V)}{\tau_h(V)}$$

Time constants are voltage dependent

The gate dynamics {n,m,h} are **fast**

→ V changes quickly (~ ms spike)

→ No slow dynamics in HH model



# Beta rhythm model

**Q.** How do we add the slow dynamics?

**A.** Add a new current

$$C \frac{dV}{dt} = \boxed{I_{Na} + I_K + I_L + I_{inj}} + \boxed{I_{new}}$$

original HH (fast)                      new current (will set the slow timescale)

**Q.** What is  $I_{new}$ ?



# Beta rhythm model

## Currents

- “fast” sodium & “fast” potassium → original HH model
- many others exist

... with V-dependent steady-states & time constants

TABLE A2. *Ionic conductance kinetic parameters*

| Conductance Type                                                 | Steady-State Activation/Inactivation                                                                      | Time Constant (ms)                                                                                                  |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| $g_{\text{Na}}$ -transient: activation                           | $1/\{1 + \exp[(-V - 34.5)/10]\};$                                                                         | $0.025 + 0.14 \exp[(V + 26.5)/10] \quad [V \leq -26.5]$<br>$0.02 + 0.145 \exp[(-V - 26.5)/10] \quad [V \geq -26.5]$ |
| $g_{\text{Na}}$ -transient: inactivation                         | $1/\{1 + \exp[(V + 59.4)/10.7]\};$                                                                        | $0.15 + 1.15/\{1 + \exp[(V + 33.5)/15]\}$                                                                           |
| $g_{\text{Na}}$ -persistent: activation                          | $1/\{1 + \exp[(-V - 48)/10]\};$                                                                           | $0.025 + 0.14 \exp[(V + 40)/10] \quad [V \leq -40]$<br>$0.02 + 0.145 \exp[(-V - 40)/10] \quad [V \geq -40]$         |
| $g_{\text{K}}$ -delayed rectifier: activation                    | $1/\{1 + \exp[(-V - 29.5)/10]\};$                                                                         | $0.25 + 4.35 \exp[(V + 10)/10] \quad [V \leq -10]$<br>$0.25 + 4.35 \exp[(-V - 10)/10] \quad [V \geq -10]$           |
| $g_{\text{K}}$ -transient (“A”): activation                      | $1/\{1 + \exp[(-V - 60)/8.5]\};$                                                                          | $0.185 + 0.5/\{\exp[(V + 35.8)/19.7] + \exp[(-V - 79.7)/12.7]\}$                                                    |
| $g_{\text{K}}$ -transient (“A”): inactivation                    | $1/\{1 + \exp[(V + 78)/6]\};$                                                                             | $0.5/ \quad [V \leq -63]$<br>$\{\exp[(V + 46)/5] + \exp[(-V - 238)/37.5]\}$<br>$9.5 \quad [V \geq -63]$             |
| $g_{\text{K}}$ -“K2”: activation                                 | $1/\{1 + \exp[(-V - 10)/17]\};$                                                                           | $4.95 + 0.5/\{\exp[(V - 81)/25.6] + \exp[(-V - 132)/18]\}$                                                          |
| $g_{\text{K}}$ -“K2”: inactivation                               | $1/\{1 + \exp[(V + 58)/10.6]\};$                                                                          | $60 + 0.5/\{\exp[(V - 1.33)/200] + \exp[(-V - 130)/7.1]\}$                                                          |
| $g_{\text{Ca}}$ -low-threshold: activation                       | $1/\{1 + \exp[(-V - 56)/6.2]\};$                                                                          | $0.204 + 0.333/\{\exp[(V + 15.8)/18.2] + \exp[(-V - 131)/16.7]\}$                                                   |
| $g_{\text{Ca}}$ -low-threshold: inactivation                     | $1/\{1 + \exp[(V + 80)/4]\};$                                                                             | $0.333 \exp[(V + 466)/66.6] \quad [V \leq -81]$<br>$9.32 + 0.333 \exp[(-V - 21)/10.5] \quad [V \geq -81]$           |
| Anomalous rectifier                                              | $1/\{1 + \exp[(V + 75)/5.5]\};$                                                                           | $1/\{\exp[-14.6 - 0.086V] + \exp[-1.87 + 0.07V]\}$                                                                  |
| Conductance Type                                                 | Forward Rate Function ( $\alpha$ )                                                                        | Backward Rate Function ( $\beta$ )                                                                                  |
| $g_{\text{K}}$ -Ca- & V-dependent (“C”) (voltage-dependent term) | $0.053 \exp[(V + 50)/11 - (V + 53.5)/27]$<br>$[V \leq -10]$<br>$2 \exp[(-V - 53.5)/27]$<br>$[V \geq -10]$ | $2 \exp[(V - 53.5)/27] - \alpha$<br>$[V \leq -10]$<br>$0$<br>$[V \geq -10]$                                         |
| $g_{\text{K}}$ -“M”                                              | $0.02/\{1 + \exp[(-V - 20)/5]\};$                                                                         | $0.01 \exp[(-V - 43)/18]$                                                                                           |
| $g_{\text{K}}$ -AHP                                              | $\min(10^{-4} \chi, 0.01);$                                                                               | $0.01$                                                                                                              |
| $g_{\text{Ca}}$ -high-threshold                                  | $1.6/\{1 + \exp[-0.072(V - 5)]\};$                                                                        | $0.02 (V + 8.9)/\{\exp[(V + 8.9)/5] - 1\}$                                                                          |

[Traub et al., J Neurophysiol, 2003]

# Beta rhythm model

Ignore details & assume 4 candidate currents

$\{A, B, C, D\}$

We know their steady-state functions  $x_{\infty}[V]$  and time constants  $\tau_x[V]$

Note voltage dependent

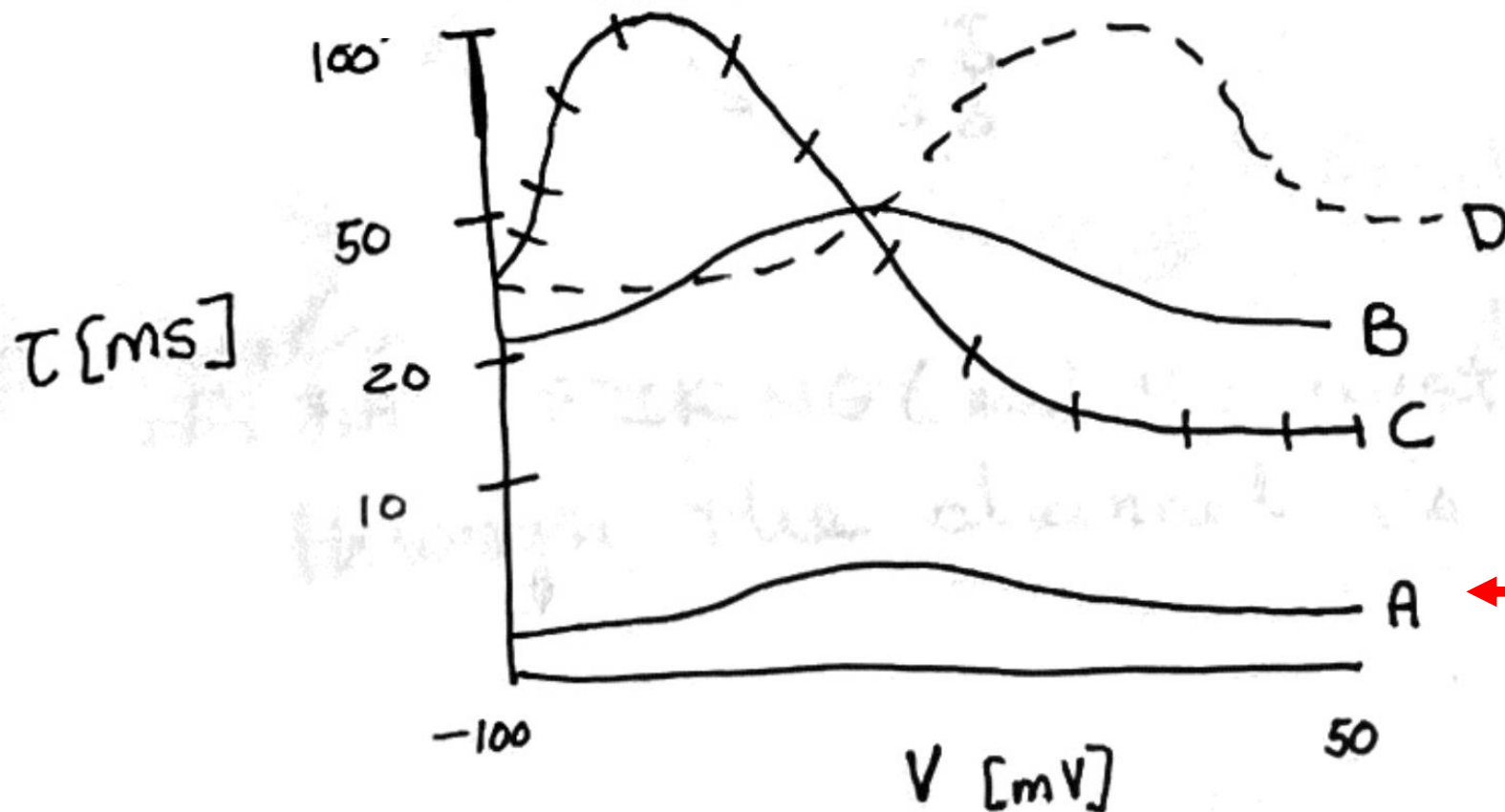
**Goal**: Determine which current could produce bursting we observe

A process of elimination ...

# Beta rhythm model

~~A~~ B, C, D}

Investigate the time constants  $\tau_x[V]$



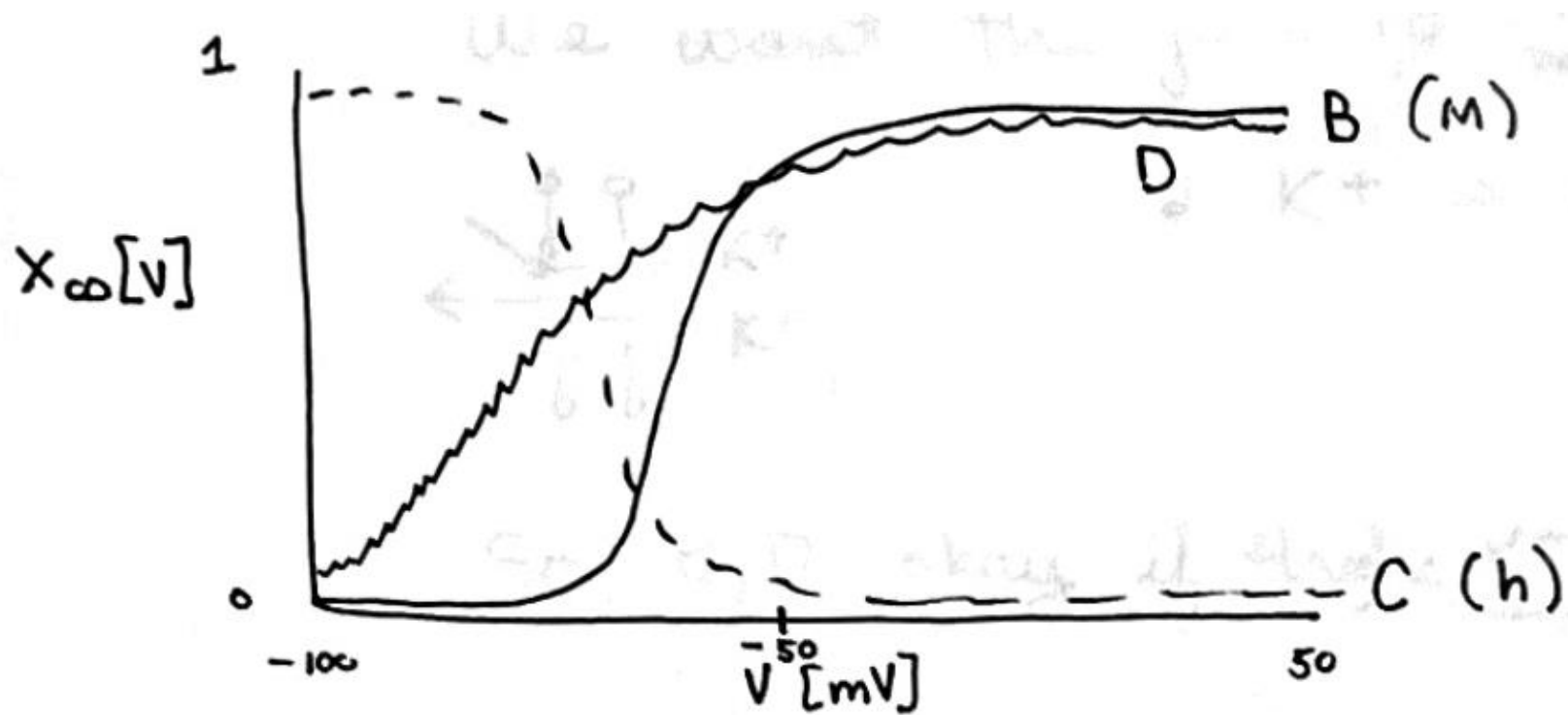
We want a slow current

- Eliminate A
- too fast for bursting

# Beta rhythm model

~~A~~ B, C, D

Investigate the steady-state functions  $x_\infty[V]$ .

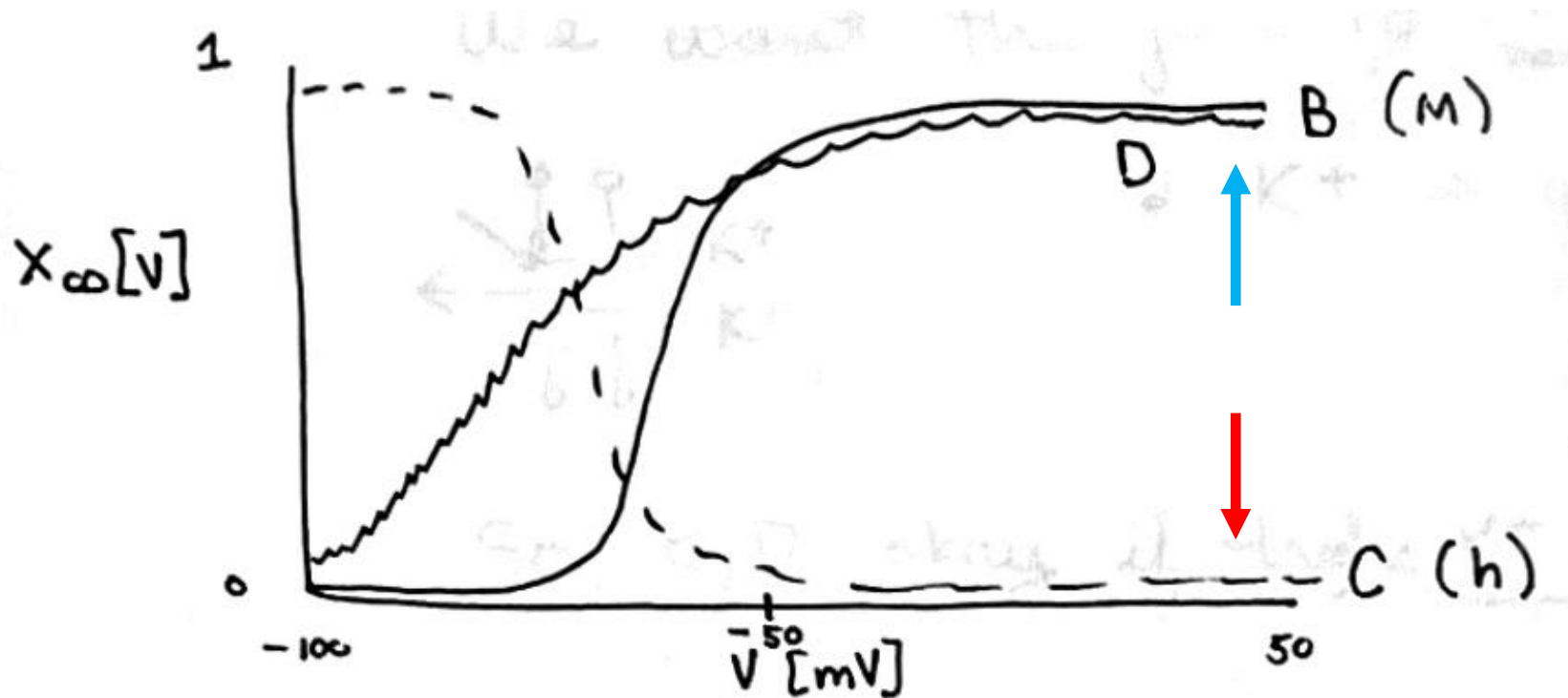


Q. Which one?

# Beta rhythm model

~~{A, B, C, D}~~

Two cases: Gate changes (slowly) during the burst



burst  $\rightarrow$  a depolarized state

Gates: (B, D) “depolarization activated”  
(C) “depolarization inactivated”

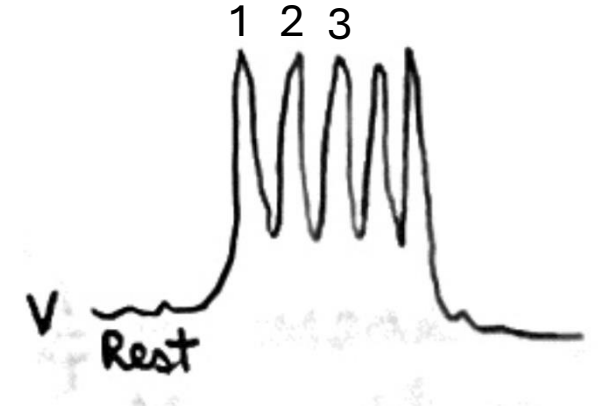
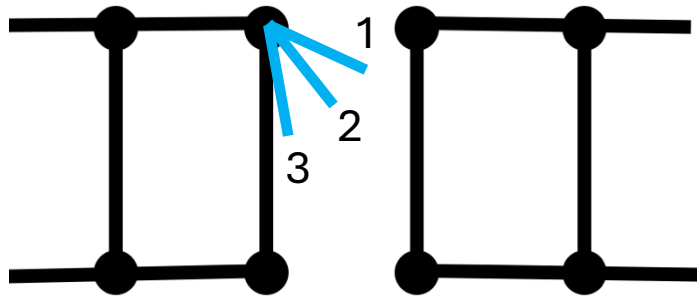
gates **open** during burst  
gates **close** during burst

# Beta rhythm model

~~A~~ B, C, D

Case 1: Gate **opens** (slowly) during the burst (B,D)

Each spike opens the gate a little bit



burst → a depolarized state

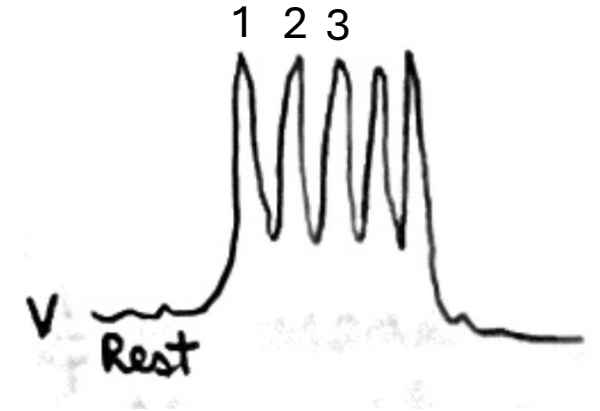
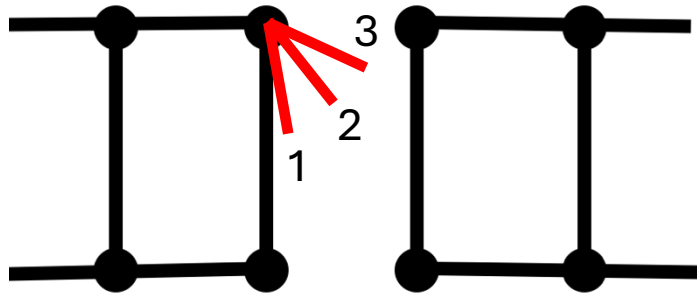
when gate is open enough, this current halts the spiking

# Beta rhythm model

~~{A, B, C, D}~~

Case 2: Gate **closes** (slowly) during the burst (C)

Each spike closes the gate a little bit



burst  $\rightarrow$  a depolarized state

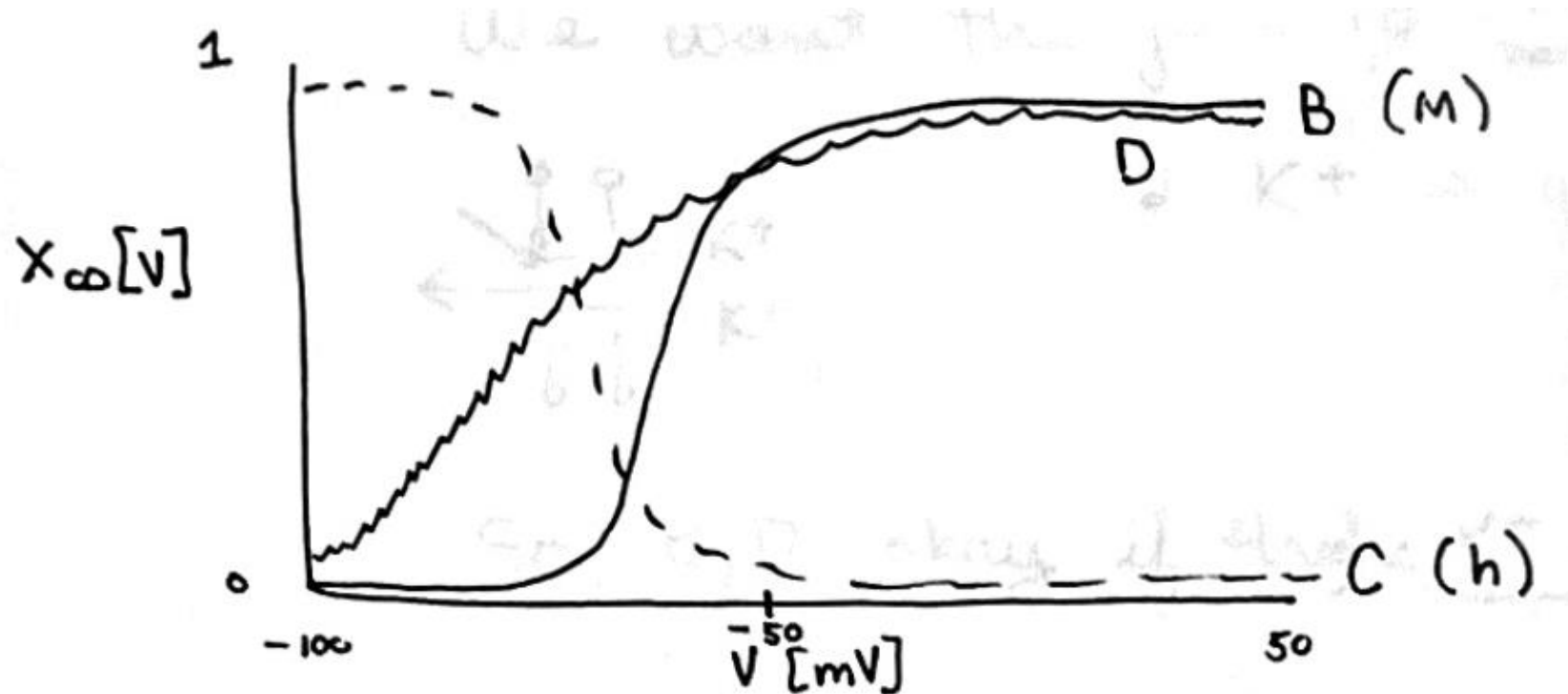
when gate is closed enough, this current halts the spiking

# Beta rhythm model

~~{A, B, C, D}~~

Case 1: Gate **opens** (slowly) during the burst (B,D)

Case 2: Gate **closes** (slowly) during the burst (C)



In either case, to halt spiking (end the burst), the type of ion that flows through the channel is critical.



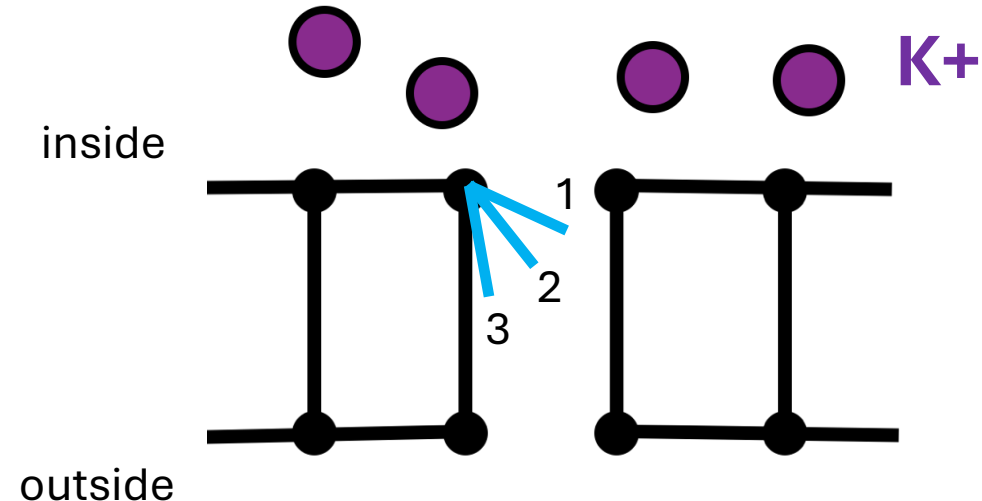
# Beta rhythm model

~~{A, B, C, D}~~

Case 1: Gate **opens** (slowly) during the burst (B,D)

Q. What ion is gated?

We want the gate to open & halt bursting



Gate opens enough  
K<sup>+</sup> released from cell  
hyperpolarize neuron

So (B,D) work if they're K<sup>+</sup> currents

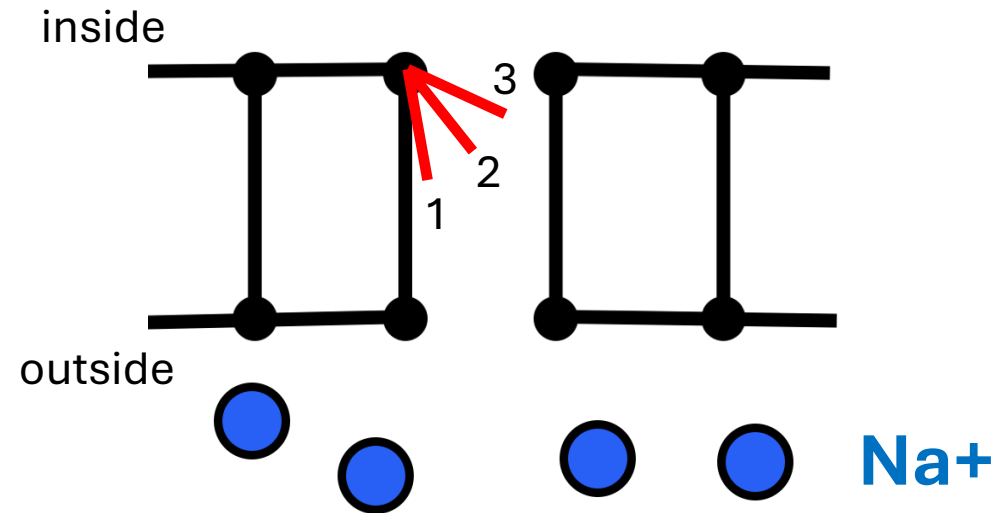
# Beta rhythm model

~~A~~ B, C, D}

Case 2: Gate **closes** (slowly) during the burst (C)

Q. What ion is gated?

We want the gate to close & halt bursting

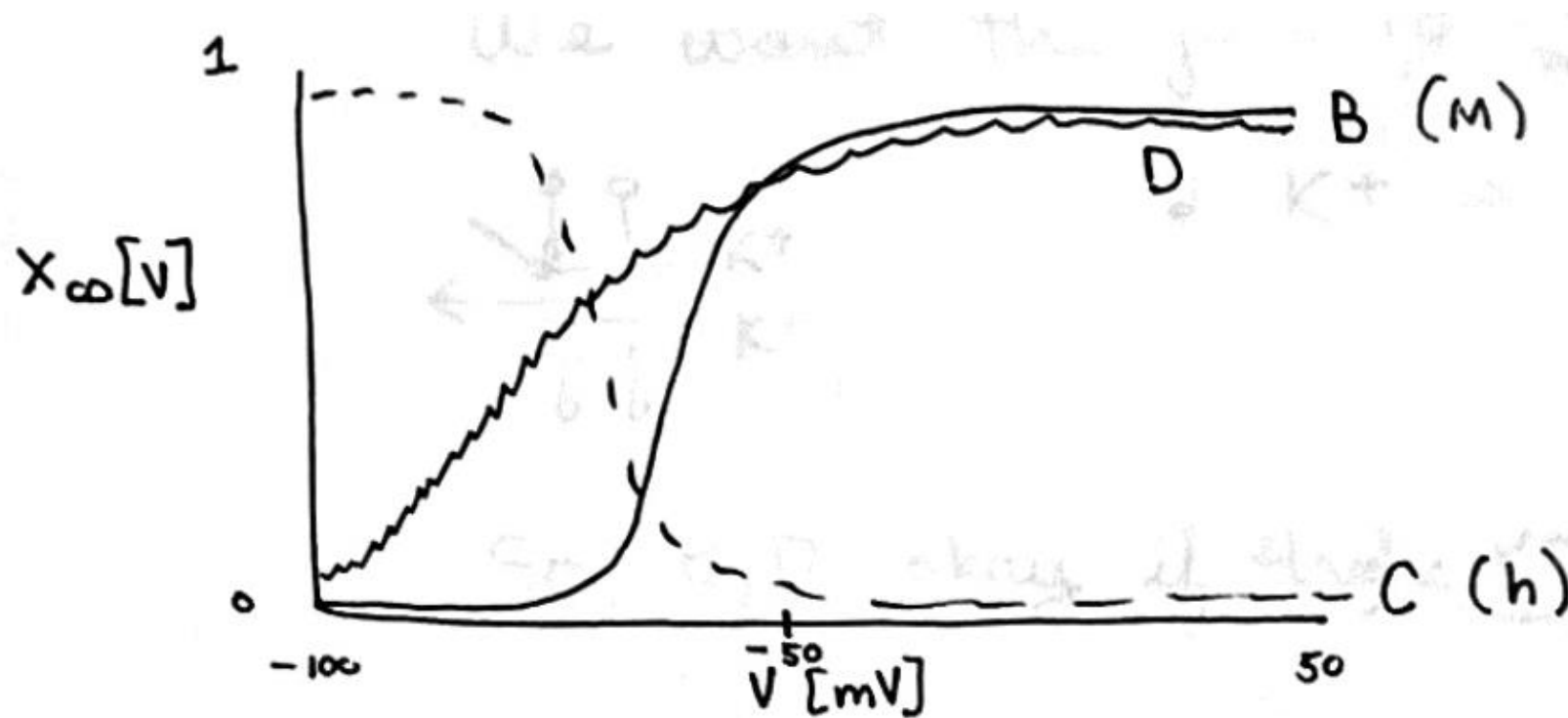


Gate closes enough  
Na<sup>+</sup> cannot enter cell  
hyperpolarize neuron

So (C) works if it's Na<sup>+</sup> (or Ca<sup>2+</sup>) currents

# Beta rhythm model

## Summary



(B,D)

okay if K<sup>+</sup> current  
slowly open & halt burst

(C)

okay if Na<sup>+</sup> current  
slowly close & halt burst

~~{A, B, C, D}~~  
↑

We learn:

B gates K<sup>+</sup>



C gates Na<sup>+</sup>



D gates Na<sup>+</sup>



~~A~~, B, C, ~~D~~

# Beta rhythm model

Two options remain for  $I_{new}$

**B**: slow, depolarization activated, outward K<sup>+</sup> current

**C**: slow, depolarization inactivated, inward Na<sup>+</sup> current

Sketch a model for **B**

# Beta rhythm model

~~A~~ B, C, ~~D~~

HH + B

$$\begin{aligned} C \frac{dV}{dt} &= I_{\text{input}}(t) - \bar{g}_K n^4 (V - V_K) - \bar{g}_{\text{Na}} m^3 h (V - V_{\text{Na}}) - \bar{g}_L (V - V_L) - \boxed{g_B B (V - E_K)} \\ \frac{dn}{dt} &= -\frac{n - n_\infty(V)}{\tau_n(V)} \\ \frac{dm}{dt} &= -\frac{m - m_\infty(V)}{\tau_m(V)} \\ \frac{dh}{dt} &= -\frac{h - h_\infty(V)}{\tau_h(V)} \end{aligned}$$

linear (from lit.)      K+ current

$$\frac{dB}{dt} = \frac{B_\infty[V] - B}{\tau_B[V]}$$

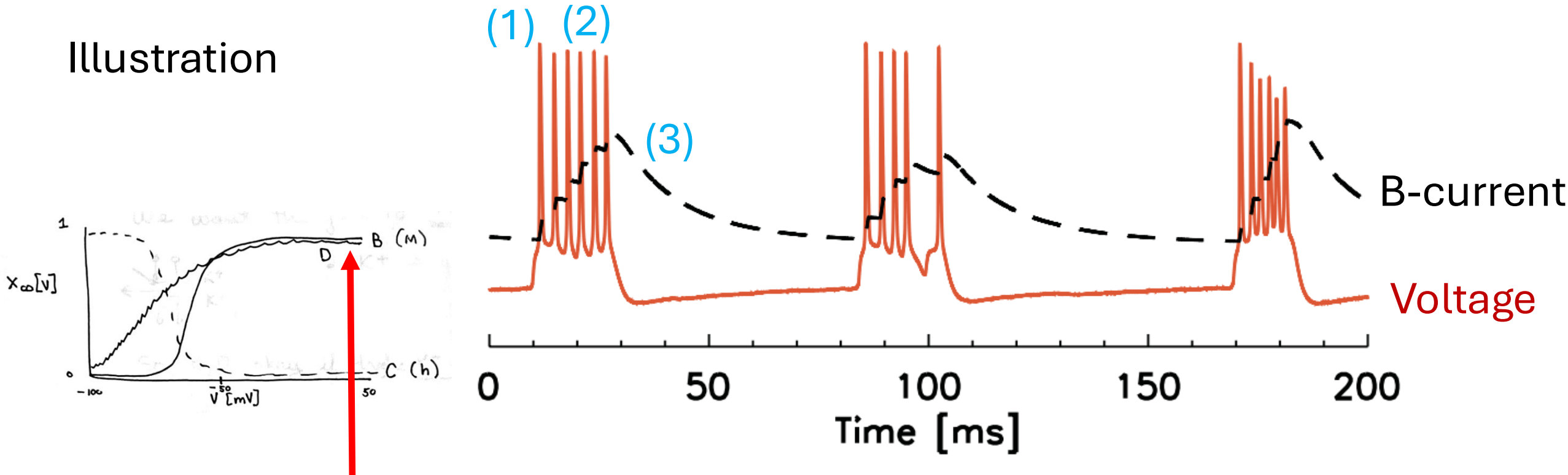
 where  $B_\infty[V]$  and  $\tau_B[V]$  plotted above

Note: model now has 5 variables {V, n, m, h, B}

# Beta rhythm model

~~B~~, B, ~~C~~, ~~D~~

Illustration



(1) During spike,  $B_{\infty}[V] \rightarrow 1$  so,  $B \rightarrow 1$  but slowly so, B increases a little

(2) repeat for multiple spikes  $B \rightarrow 1$  B opens enough to halt spiking

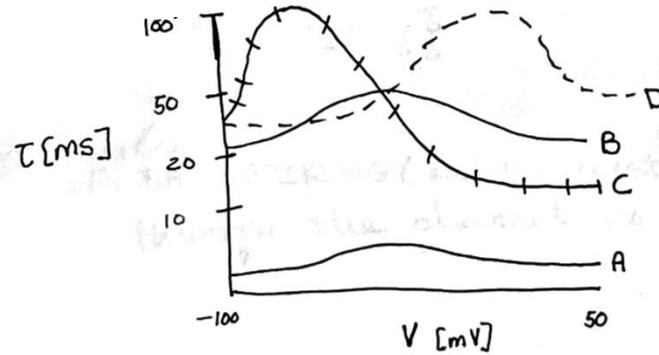
(3) B decays  $\rightarrow 0$  another burst starts

# Summary

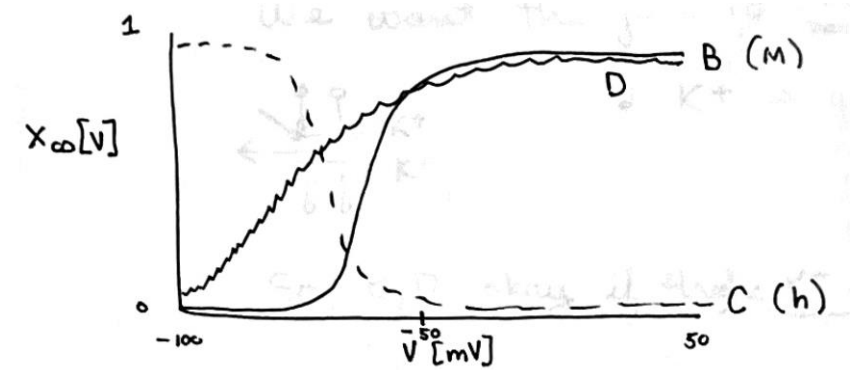
A procedure to classify the dynamic effects of a current on a neuron

## 3 questions

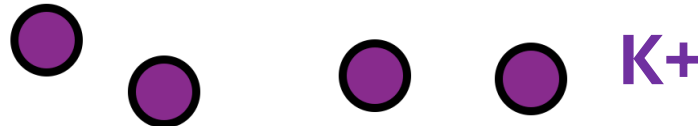
1. Is it fast or slow?



2. Is it depolarization activated or inactivated?



3. What ion is gated?



Then, we know enough to predict dynamic behavior.

# Beta rhythm model



*Python*